

Candidates' Performance

Paper 1

Paper 1 consisted of two sections: Section A (multiple-choice questions) and Section B (conventional questions). Sections A and B each comprised two parts: Part I and Part II. Part I contained questions mainly on Topics I to VIII of the curriculum, while Part II contained questions mainly on Topics IX to XII. All questions in both sections were compulsory.

Section A (multiple-choice questions)

This section consisted of 36 multiple-choice questions. The mean score was 24.5. The performance of candidates was generally good. Some misconceptions were revealed by candidates in their performance on the following items:

- For Q.9, it should be noted that the solubility of an organic compound in water depends on the nature of the functional group and the length of hydrophobic hydrocarbon chain. In this item, compound Y has two polar hydroxyl ($-\text{OH}$) groups and a short hydrocarbon chain, therefore it can form strong hydrogen bonds with water molecules and it has the highest solubility in water. Compound Z has a polar ester ($-\text{COO}-$) group and a short hydrocarbon chain, therefore it can form hydrogen bonds with water molecules and it is water soluble. Although compound X contains one polar hydroxyl ($-\text{OH}$) group, it has the longest hydrocarbon chain among the three molecules, therefore it has the weakest attraction with water molecules and it has the lowest solubility in water.

Q.9 Consider the following three compounds :



Which of the following shows the decreasing order of their solubilities in water ?

- | | | |
|-----|-----------|-------|
| A. | X > Y > Z | (5%) |
| B. | Z > Y > X | (9%) |
| C.* | Y > Z > X | (20%) |
| D. | Y > X > Z | (66%) |

- For Q.18, copper is a rather unreactive metal and there is no reaction between copper and concentrated $\text{HCl}(\text{aq})$. Thus, statement (1) is wrong. Iron reacts slowly with dilute sulphuric acid to form hydrogen gas. Thus, statement (2) is correct. Many candidates were not aware that calcium is a reactive metal and it can react with water in $\text{NaOH}(\text{aq})$ to form hydrogen gas. Thus, statement (3) is also correct.

Q.18 Which of the following pairs of substances, when mixed, would release hydrogen gas ?

- | | | |
|-----|---|-------|
| (1) | copper and concentrated $\text{HCl}(\text{aq})$ | |
| (2) | iron and $\text{H}_2\text{SO}_4(\text{aq})$ | |
| (3) | calcium and $\text{NaOH}(\text{aq})$ | |
| | | |
| A. | (1) only | (13%) |
| B. | (2) only | (64%) |
| C. | (1) and (3) only | (4%) |
| D.* | (2) and (3) only | (19%) |

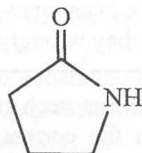
3. For Q.22, about half of the candidates correctly chose the answer D. Candidates should note that the pH of an acid depends on the concentration and the strength of the acid. In this item, acid **A** and acid **B** have the same basicity and concentration, but **A(aq)** has a lower pH than **B(aq)**. This means that **A** is a stronger acid than **B**, **A** has a greater extent of ionisation and **A** gives a higher concentration of $\text{H}^+(\text{aq})$ ions. Therefore, statement (1) is correct. Since **B** is a weaker acid, some **B** molecules do not ionise and are present in **B(aq)**. Therefore, statement (2) is correct. For statement (3), since both the volumes and concentrations of **A(aq)** and **B(aq)** are the same, the respective total number of moles of ionisable hydrogen atoms present in **A(aq)** and **B(aq)** is also the same. Thus, the same number of moles of NaOH(aq) is required for the respective complete neutralisation of **A(aq)** and **B(aq)**. Therefore, statement (3) is also correct.

Q.22 Both **A** and **B** are monobasic acids. The pH of 0.10 M **A(aq)** is 1.0 and the pH of 0.10 M **B(aq)** is 3.0. Which of the following statements are correct ?

- | | | |
|-----|---|-------|
| (1) | A is a stronger acid than B . | |
| (2) | Some B molecules are present in B(aq) . | |
| (3) | Complete neutralisation of 25.0 cm ³ of 0.10 M A(aq) and complete neutralisation of 25.0 cm ³ of 0.10 M B(aq) require the same number of moles of NaOH(aq) . | |
| A. | (1) and (2) only | (24%) |
| B. | (1) and (3) only | (16%) |
| C. | (2) and (3) only | (6%) |
| D.* | (1), (2) and (3) | (54%) |

4. For Q.29, about a third of the candidates wrongly chose option D as the answer. Many candidates neglected the effect of acidic and alkaline media on a chemical reaction. In this item, the amide group in the given organic compound undergoes alkaline hydrolysis in excess NaOH(aq) . A -COO^- ion is formed at one end of the carbon chain and a -NH_2 group is formed at the other end. After the addition of excess HCl(aq) , the -COO^- ion is converted into a -COOH group while the -NH_2 group is converted into $\text{-NH}_3^+\text{Cl}^-$. Thus, B is the correct answer.

Q.29 The structure of an organic compound is shown below :



When it is heated with excess NaOH(aq) , followed by the addition of excess HCl(aq) , a major organic product **Z** is formed. Which of the following is **Z** ?

- | | | | |
|----|-------|-----|-------|
| A. | | B.* | |
| | (15%) | | (40%) |
| C. | | D. | |
| | (12%) | | (33%) |

Section B (conventional questions)

Question Number	Performance in General
1	The performance of candidates was very good. In part (a), a very high percentage of the candidates were able to give the correct answer 'isotopes'. A small number of the candidates misspelt the word 'isotopes', or wrongly gave 'isomers' as the answer. In part (b), about three quarters of the candidates were able to recognise that x is 18. A small number of candidates wrongly gave '8' as the answer. In part (c), about three quarters of the candidates were able to draw the correct electron diagram for potassium iodide. A small number of the candidates wrongly considered potassium iodide to be a covalent compound and drew an incorrect electron diagram. A few candidates gave incorrect numbers of the outermost shell electrons for K^+ and I^-. In part (d), about three quarters of the candidates were able to state that an aqueous solution of hydrogen iodide has mobile ions to conduct electricity. A small number of candidates wrongly stated that hydrogen iodide is an ionic compound, or an aqueous solution of hydrogen iodide has delocalised electrons to conduct electricity. In part (e), about a third of the candidates were able to state that potassium iodide has a higher melting point, and give a correct and complete explanation. A small number of candidates wrongly stated that both potassium iodide and hydrogen iodide have giant ionic structures, or hydrogen iodide has a giant covalent structure. Some candidates wrongly wrote that strong ionic bonds exist between potassium iodide molecules.
2	The performance of candidates was very good. In part (a), a very high percentage of the candidates were able to state that the colourless gas Z is oxygen, and about three quarters of the candidates were able to correctly state that oxygen can relight a glowing splint. A small number of candidates wrongly considered that the colourless gas Z is hydrogen or carbon dioxide, and suggested an incorrect corresponding test. In part (b)(i), about two thirds of the candidates were able to calculate the relative atomic mass of X correctly. A small number of candidates were confused about the stoichiometry among X, X_2O and O_2. Some candidates wrongly put an inappropriate unit (e.g. $g\ mol^{-1}$) for the relative atomic mass. In part (b)(ii), about three quarters of the candidates were able to correctly state that X is silver. In part (c), about two thirds of the candidates were able to answer this question correctly. Many students were able to correctly explain why the reaction is a redox reaction in terms of change in oxidation numbers. Some candidates were not able to give the oxidation numbers of the species precisely, for example, they wrongly gave +2 as the oxidation number of X in X_2O.
3	The performance of candidates was very good. In part (a), a very high percentage of the candidates were able to write the correct chemical equation for the reaction between $NaHCO_3(s)$ and $HCl(aq)$. A very small number of candidates failed to balance the equation correctly. In part (b), about three quarters of the candidates were able to calculate the percentage by mass of $NaHCO_3(s)$ in the antacid sample correctly. A few candidates made an inappropriate round-off in the calculation and gave an incorrect numerical value for the percentage. In part (c)(i), just below half of the candidates were able to recognise that methyl orange is a suitable indicator for the titration and state the correct colour change at the end point of the titration. A small number of candidates wrote the wrong colour change, like 'yellow to red' or 'orange to yellow'. A few candidates wrongly suggested that phenolphthalein is a suitable indicator for the titration. In part (c)(ii), about three quarters of the candidates were able to suggest a suitable instrument to measure the pH of the solution accurately. Common mistakes in the answers included 'universal indicator', 'pH paper' and 'colorimeter'. In part (d), about half of the candidates were able to correctly state an advantage of taking antacids containing $Mg(OH)_2(s)$ over those containing $NaHCO_3(s)$. Common mistakes in the answers included '$NaHCO_3(s)$ is more soluble in water than $Mg(OH)_2(s)$', 'the reactivity of Mg is lower than that of Na' and '$Mg(OH)_2(s)$ can react with more $HCl(aq)$ than $NaHCO_3(s)$'.

Question Number	Performance in General
4	The performance of candidates was satisfactory. In part (a)(i), about half of the candidates were able to recognise that a H_3O^+ ion is formed from a water molecule and a hydrogen ion, and draw the correct electron diagrams in the answer. Some candidates wrongly drew an electron diagram of an oxide ion instead of a water molecule, or an electron diagram of a hydrogen molecule or a hydrogen atom instead of a hydrogen ion. In part (a)(ii), about a third of the candidates were able to accurately describe the formation of the dative covalent bond as the sharing of a lone pair on the oxygen atom of a water molecule with a hydrogen ion. A small number of the candidates wrongly stated that the dative covalent bond is formed between a water molecule and a hydrogen atom. In part (b), about half of the candidates were able to correctly state that the boron atom in a BF_3 molecule does not have an octet structure because it has only six electrons in its outermost shell. A small number of the candidates wrongly considered that a boron atom has five outermost shell electrons before the formation of the B–F bonds. In part (c)(i), about two thirds of the candidates were able to draw the correct three-dimensional structure of a SF_6 molecule. Some candidates were confused about the directions of the six covalent bonds in the structure. In part (c)(ii), about half of the candidates were able to state that SF_6 is a non-polar molecule and give a correct explanation. Common mistakes in the answers included ‘SF_6 molecule has a trigonal bipyramidal structure’, ‘the electronegativities of the F atoms cancelled out each other’ and ‘the polar S–F bonds cancelled out each other’. In part (d), about a third of the candidates were able to correctly explain for the increasing order of the boiling points of BF_3, SF_6, and H_2O, in terms of the nature and the comparative strength of intermolecular forces. Some candidates failed to mention that hydrogen bonds are stronger than van der Waals’ forces. A small number of candidates wrongly mentioned that the van der Waals’ forces between SF_6 molecules are stronger than those between BF_3 molecules because the molecular mass instead of molecular size of SF_6 is larger.
5	The performance of candidates was satisfactory. In part (a), about three quarters of the candidates were able to give the meaning of ‘primary cell’. In part (b)(i), fewer than half of the candidates were able to write the required half equation. Quite a number of candidates did not include $\text{OH}^-(\text{aq})$ ions in the equations. In part (b)(ii), only about a quarter of the candidates were able to suggest the disadvantage of using this kind of hydrogen-oxygen fuel cell, like ‘hydrogen is flammable’. In parts (c)(i) and (ii), about two thirds of the candidates were able to write the required equations. In part (c)(iii), about one third of the candidates were able to suggest how to obtain aluminium from aluminium oxide. Some candidates did not include ‘molten’ in their answers.
6	The performance of candidates was good. In part (a), about three quarters of the candidates were able to write the name of the type of reaction involved. A small number of candidates gave the wrong name, like ‘addition reaction’, and some candidates were not able to correctly spell ‘substitution’. In part (b), a very high proportion of the candidates were able to suggest the reaction condition, like ‘diffused sunlight’ or ‘ultra-violet’. In part (c)(i), just over half of the candidates were able to explain why a chlorine free radical is reactive using terms such as ‘the presence of an unpaired electron’, but some candidates just mentioned ‘7 outermost shell electrons’ without further elaboration. In part (c)(ii), just over half of the candidates were able to state the chemical species involved in the different stages of the substitution reaction. In part (d), only a small number of the candidates were able to use ‘further substitution’ or similar terms to explain why CH_3Cl is not the only organic product. In part (e), a very high proportion of the candidates were able to correctly select the ‘flammable’ hazard warning label.

Question Number	Performance in General
7	The performance of candidates was fair. In part (a), a very high proportion of the candidates were able to write the equation for the reaction between $\text{Ca(OH)}_2(\text{s})$ and $\text{HCl}(\text{aq})$. In part (b)(i), just under half of the candidates were able to determine the change in temperature from the line graph. Some candidates were not able to correctly interpret the meaning of 'the greatest temperature rise'. In part (b)(ii), about half of the candidates were able to calculate the required enthalpy change of neutralisation. In part (c), only a very small number of the candidates were able to calculate the standard enthalpy change of the given reaction. A very high proportion of the candidates did not recognise that the enthalpy change of the neutralisation refers to the energy change during the formation of one mole of water under certain temperature and pressure.
8	The performance of candidates was fair. About half of the candidates were able to describe the similarities of the two rust prevention methods. However, some candidates did not mention that a complete metal coating on the iron-made object is required to prevent iron from coming into contact with oxygen and water. About half of the candidates were able to describe the differences between the two rust prevention methods using appropriate chemical principles. Some candidates were able to present relevant concepts logically using sentences like 'when the zinc coating is scratched, as the reactivity of zinc is higher than that of iron, zinc will lose electrons more readily and thus the galvanised iron will not rust'. Some candidates mixed up 'anodisation', 'cathodic protection' and 'galvanisation'.
9	The performance of candidates was good. In part (a)(i), about two thirds of the candidates were able to calculate the reaction quotient of the system correctly and give the answer with a correct unit. A very high percentage of the candidates were able to give the correct expression of Q_c in terms of the concentrations of the species, but some of them failed to notice that the volume of reaction mixture was 4.0 dm^3, and so they omitted this data in the calculations. A very small number of the candidates used the 'number of moles' of the species instead of concentrations in the calculations. Some candidates gave an incorrect unit in their answers. In part (a)(ii), about half of the candidates were able to recognise that Q_c was smaller than K_c when the reactants were initially introduced into the closed container, and correctly stated that the concentration of $\text{PCl}_5(\text{g})$ would decrease just after the reaction started. A small number of candidates failed to compare the values of Q_c and K_c, and wrongly considered that the relative rates of the forward and backward reactions are simply governed by the amounts of the reaction species in the system. In part (b), about half of the candidates were able to correctly deduce that K_c would increase if the temperature of the equilibrium mixture is increased and give a complete explanation. Some candidates failed to give a proper explanation. For example, some candidates failed to mention that the forward reaction is endothermic and they merely stated that the equilibrium would shift to the right when the temperature is increased. A small number of candidates wrongly stated that change of temperature would only affect the position of an equilibrium but not the value of K_c.

Question Number	Performance in General
10	The performance of candidates was satisfactory. In part (a), about three quarters of the candidates were able to give the correct chemical equation for the decomposition of $\text{H}_2\text{O}_2(\text{aq})$. A small number of candidates failed to balance the equation correctly. A few candidates wrongly gave a 'reversible' sign for the chemical equation. In part (b), about a third of the candidates were able to correctly state that manganese exhibits catalytic properties because the reaction rate increased upon the addition of $\text{MnO}_2(\text{s})$ to the reaction mixture. A small number of candidates merely stated that $\text{MnO}_2(\text{s})$ remained chemically unchanged at the end of the reaction without mentioning the effect of $\text{MnO}_2(\text{s})$ on the reaction rate. Some candidates wrote incorrect chemical species in their answers, such as 'Mn^{2+}'. In part (c), about two thirds of the candidates were able to correctly calculate the theoretical volume of $\text{O}_2(\text{g})$ released upon completion of the reaction as being 0.36 dm^3. A small number of candidates failed to recognise that the mole ratio of $\text{H}_2\text{O}_2(\text{aq})$ and $\text{O}_2(\text{g})$ in the reaction is 2:1, and so they gave an incorrect result in calculation. In part (d), about one third of the candidates were able to give a correct and complete explanation. About two thirds of the candidates were able to state that the concentration of hydrogen peroxide decreased as the reaction proceeded, resulting in a decrease in reaction rate. Some candidates failed to mention that the time taken to produce foam became longer at the later reaction stage. Some wrong answers included 'the "amount" of reactant decreased' and '$\text{MnO}_2(\text{s})$ was used up during the experiment'.
11	The performance of candidates was satisfactory. In part (a), about half of the candidates were able to give the correct systematic name for compound P. Some candidates gave incorrect numbers to indicate the positions of the methyl group and the hydroxyl group, like '1-methylbutan-2-ol' or '2-methylbutan-1-ol', and some candidates gave the name written as two separated parts, such as '2-methyl butan-2-ol'. In part (b)(i), about a third of the candidates were able to draw a correctly labelled diagram for a reflux set-up. Many candidates were not able to draw a proper set-up diagram even though they were able to give the correct labels. Common mistakes in the answers included 'having a thermometer or a glass stopper put at the top of the water condenser', 'the positions of water in and water out on the condenser were incorrect' and 'there was a large gap drawn between the water condenser and the flask'. A few candidates wrongly drew a set-up diagram of simple distillation or fractional distillation. In part (b)(ii), about three quarters of the candidates were able to correctly state that the expected colour change of the reaction mixture is from orange to green. In part (b)(iii), about half of the candidates were able to give the correct structural formula of the organic product obtained from the oxidation of Q. A small number of candidates failed to recognise that the product is a ketone, and wrongly gave the structure of a carboxylic acid or an aldehyde. Some candidates gave the structure of a ketone with an incorrect carbon number. A small number of candidates wrongly drew an extra hydrogen atom attaching to the carbon atom of $\text{C}=\text{O}$ group. In part (c)(i), about two thirds of the candidates were able to give a correct structural formula for compound W. A few candidates wrongly added an extra CH_2 group in the carbon chains. In part (c)(ii), about half of the candidates were able to suggest a suitable reducing agent required for the reaction. Some candidates wrongly gave '$\text{LiAlH}_4 / \text{H}^+(\text{aq})$' or '$\text{LiAlH}_4(\text{aq})$' in their answers. In part (d), only about a quarter of the candidates were able to give the correct structural formula for compound S. Many candidates failed to recognise that compound S is a secondary alcohol. There were also quite a number of candidates who merely gave the structural formula of Q in their answers and failed to recognise that they should give the structure of an isomer of Q.

Question Number	Performance in General
12	The performance of candidates was good. Just below half of the candidates were able to suggest a correct and complete synthetic route for converting butane-1,4-diol to butane, and give the correct reagents and conditions required for each step. Some candidates failed to state the reagent and condition required in the first step precisely. Common mistakes in the answers included '$\text{H}_2\text{SO}_4(\text{aq})$', '$\text{conc. H}_2\text{SO}_4(\text{aq})$', 'without stating H_2SO_4 is concentrated' and 'heating is omitted'. Some candidates forgot to write the catalyst for the hydrogenation reaction in the second step. A small number of candidates drew the structure of the reaction intermediate with an incorrect carbon chain length.
13	The performance of candidates was fair. Only a small number of candidates were able to give a correct and complete answer. A small number of candidates did not read the question carefully and failed to notice that they were asked to describe the acid-base properties of the 'products' formed when the oxides are added into water. They did not mention any of the 'products' formed in the reactions, and only stated the acid-base properties of the oxides. About two thirds of the candidates were able to correctly state that alkaline solution of sodium hydroxide will be formed when Na_2O is added to water. Some candidates wrongly considered that MgO is not soluble in water or it does not react with water. About half of the candidates were able to state that Al_2O_3 is insoluble in water or it does not react with water, so that the acid-base property cannot be shown. A small number of candidates wrongly stated that alkaline aluminium hydroxide will be formed when Al_2O_3 is added to water. Even though about half of the candidates were able to recognise that an acidic product will be formed when Cl_2O is added to water, many of them failed to state the correct product. Some candidates confused the reaction of Cl_2 and water with the reaction of Cl_2O and water, thus they wrongly stated that a mixture of $\text{HCl}(\text{aq})$ and $\text{HOCl}(\text{aq})$ would be formed.

Paper 2

Paper 2 consisted of three sections. Section A contained questions set on Topic XIII 'Industrial Chemistry', Section B on Topic XIV 'Materials Chemistry' and Section C on Topic XV 'Analytical Chemistry'. Candidates were required to attempt all questions in two of the sections.

Question Number	Popularity (%)	Performance in General
Section A: 1	98	The performance of candidates in part (a) was satisfactory. In part (a)(i)(1), a very high percentage of the candidates were able to suggest a reason for the reaction to be considered as green. In part (a)(i)(2), about half of the candidates were able to suggest a reason for the reaction not to be considered as green. Some candidates mentioned that CO is harmful instead of toxic as the reason. In part (a)(ii)(1), about half of the candidates were able to suggest that using a porous structure in the catalytic converters can increase the effectiveness of the catalyst by increasing its surface area. In part (a)(ii)(2), about a third of the candidates were able to state that the catalyst can be poisoned by impurities after prolonged use. Some candidates wrongly mentioned that the catalyst would be poisoned by the reactants. In part (a)(iii), about two thirds of the candidates were able to state that glass bottle is not manufactured from petrochemicals. The performance of candidates in part (b) was fair. In part (b)(i), about two thirds of the candidates were able to state that X is water. In part (b)(ii)(1), a very high proportion of the candidates were able to state that A is chlorine. In part (b)(ii)(2), about a quarter of the candidates were able to explain that chlorine is formed is due to the concentration of chloride ions being much higher than that of hydroxide ions. Some candidates only mentioned that the concentration of chloride ions is high without making a comparison. In part (b)(iii)(1), a high proportion of the candidates were able to write a half equation for the formation of B. In part (b)(iii)(2), about a third of the candidates were able to explain why sodium hydroxide is formed without containing sodium chloride. Many candidates did not mention that the chloride ions cannot pass through the membrane in explaining why there is no sodium chloride present. In part (b)(iv), about a third of the candidates were able to suggest a chemical that can be manufactured from the reaction between chlorine and sodium hydroxide solution. The performance of candidates in part (c) was fair. In part (c)(i), a high proportion of the candidates were able to state the meaning of 'initial rate'. Some candidates wrongly stated that the initial rate was the rate at the starting 'period'. In part (c)(ii), about a quarter of the candidates were able to suggest why the rate equation can be modified. Many candidates wrongly thought that the reason was the order of reaction with respect to $\text{H}^+(\text{aq})$ being zero. In part (c)(iii), about half of the candidates were able to deduce the order of reaction with respect to $\text{S}_2\text{O}_3^{2-}(\text{aq})$. Some candidates did not deduce the order of reaction with reference to the dotted lines in the graph. In part (c)(iv), about half of the candidates were able to calculate the activation energy.

Question Number	Popularity (%)	Performance in General
Section B: 2	5	The performance of candidates in part (a) was poor. In part (a)(i)(1), about one third of the candidates were able to state the meaning of the term 'nanoparticles'. In part (a)(i)(2), only a small number of the candidates were able to suggest an application of gold nanoparticles in buildings. In part (a)(ii)(1), about half of the candidates were able to correctly select the diagram that represented the expected phase of liquid crystal molecules. In part (a)(ii)(2), about one third of the candidates were able to suggest that diagram A represents the 'nematic phase' of liquid crystal molecules. In part (a)(iii), just over one third of the candidates were able to calculate the atom economy of the reaction. The performance of candidates in part (b) was poor. In part (b)(i), just over one third of the candidates were able to write the name of this type of crystal structure. In part (b)(ii), about one third of the candidates were able to deduce the number of aluminium atoms in the unit cell. In part (b)(iii), only about one fifth of the candidates were able to calculate the density of aluminium. Some candidates did not show how to calculate the volume using the given edge length. In part (b)(iv)(1), about one fifth of the candidates were able to explain why the duralumin sample has a lower melting point than pure aluminium in terms of bonding and structure. Some candidates used incorrect concepts, such as 'covalent bonding between aluminium atoms' and 'strong van der Waals' forces between aluminium molecules'. In part (b)(iv)(2), about two thirds of the candidates were able to give appropriate reasons to support their comments. The performance of candidates in part (c) was fair. In part (c)(i), about one third of the candidates were able to write the chemical equation for the polymerisation of ethene. In part (c)(ii), about half of the candidates were able to write the name for this type of polymerisation. In part (c)(iii)(1), only a small number of the candidates were able to explain why HDPE is more suitable than LDPE for making rigid plastic bottles. Some candidates did not attempt to answer using concepts, like 'HDPE chains are more linear than LDPE chains' and 'stronger van der Waals' forces exist between HDPE chains'. In part (c)(iii)(2), only about one third of the candidates were able to suggest 'blow moulding' as the appropriate moulding method. In part (c)(iv)(1), about half of the candidates were able to comment on the thermal properties of X. In part (c)(iv)(2), just over one third of the candidates were able to explain the thermal property of X from the perspective of bonding and structure. Some candidates were able to answer with appropriate terms, like 'strong cross links' and 'strong covalent bonds between polymer chains'. In part (c)(iv)(3), about half of the candidates were able to suggest another property of X. Some candidates only stated 'hardness' but did not mention 'high' or 'low'.

Question Number	Popularity (%)	Performance in General
Section C: 3	97	The performance of candidates in part (a) was fair. In part (a)(i), just below half of the candidates were able to suggest how to distinguish $\text{SO}_2(\text{g})$ and $\text{CO}_2(\text{g})$. Many candidates wrongly suggested the use of limewater and overlooked that both would have similar observations. Some candidates suggested the use of sodium dichromate or potassium permanganate solution but did not mention that acidic condition is required. In part (a)(ii), about a quarter of the candidates were able to distinguish $\text{CH}_3\text{CH}_2\text{CHO}(\text{l})$ and $\text{CH}_3\text{COCH}_3(\text{l})$ with reference to their respective mass spectra. Some candidates mentioned the specific peak in the mass spectrum of one species but did not mention whether this peak would be absent in the mass spectrum of the other species. In part (a)(iii), a high proportion of the candidates were able to state that anhydrous sodium sulphate is the most suitable for drying ethyl butanoate. The performance of candidates in part (b) was satisfactory. In part (b)(i), about two thirds of the candidates were able to show that all of Y in the solid sample should have dissolved by calculation. In part (b)(ii), about half of the candidates were able to state that the reason for using filtration is to remove the insoluble activated charcoal. In part (b)(iii), about half of the candidates were able to name the process of formation of Y(s). In part (b)(iv), just under half of the candidates were able to suggest a reason why the mass of Y(s) collected in Step (4) was smaller than the mass of Y in the solid sample. Some candidates mentioned that Y(s) will be lost without indicating the specific step involved, such as filtration. In part (b)(v)(1), just under half of the candidates were able to sketch a labelled chromatogram of TLC. Some candidates did not draw the starting position. In part (b)(v)(2), about two thirds of the candidates were able to state that the first-collected fraction is Y and gave the reason with reference to the R_f value of Y. The performance of candidates in part (c) was fair. In part (c)(i)(1), about a third of the candidates were able to state the chemical species responsible for the colour change at the end point. Some candidates wrongly mentioned that the pink colour was due to Mn^{2+} ions. In part (c)(i)(2), about two thirds of the candidates were able to calculate the concentration of $\text{Fe}^{2+}(\text{aq})$ in solution S. In part (c)(ii)(1), just below half of the candidates were able to sketch a labelled calibration curve and draw two lines on the calibration curve to show how the required concentration can be found. In part (c)(ii)(2), about a quarter of the candidates were able to suggest how the concentration of $\text{Fe}^{2+}(\text{aq})$ in solution S can be determined using the dilution factor mentioned in Step (3).